Academic Report for the Human-Agent Negotiation League - ANAC 2026

LastOffer

Leibniz Universität Hannover
Felix Bieber, Katherina Pukhova

Abstract

The `LastOfferNegotiator` is a hybrid agent combining a mathematical decision engine and a conflict-based opponent model to execute time-based, Boulware-style concessions while proposing offers structurally similar to the opponent's bids. It leverages an LLM to dynamically generate natural, argument-driven text responses.

1 Mathematical Decision Engine

1.1 Bidding Strategy

The `LastOfferNegotiator` employs a time-dependent, Boulware-style concession mechanism to calculate its target utility based on relative time. To prevent erratic concession behavior, candidate offers are strictly constrained by an upper utility bound, which is defined as the minimum between the agent's historically lowest proposed utility and a given percentage margin above the current aspiration level.

Within this defined utility window, the agent generates novel proposals whenever possible. Furthermore, in order to be perceived cooperative towards a human subject, the agent avoids offering his absolute best offer.

Additionally, to prevent exploitation, our aspiration function is fixed after half the time has passed, if the opponent has not conceded a given amount.

When selecting the proposal from the candidates, the agent evaluates each offer using a scoring heuristic. Utilizing the estimated issue weights derived from the opponent model, the agent calculates the similarity between its candidate offers and both the opponent's initial and most recent bids. The weighting of this heuristic shifts over the course of the negotiation: early rounds prioritize the maximization of the agent's own utility, whereas later stages shift focus toward maximizing similarity with the opponent's offers to reach an agreement.

Finally, as an end-game strategy, the agent reviews the opponent's offer history, reproposing them if they satisfy the agent's current aspiration threshold.

1.2 Acceptance Strategy

As the main acceptance rule the `LastOfferNegotiator` accepts every offer if it is not less than his aspiration function at the given time or if the agent has already proposed this offer before. The agent keeps track of the best opponent's offer and reject an offer if it is significantly below it as we assume a concession. As an end-game strategy the agent accepts every offer which is above his safety floor or a given percentage margin above his reservation value if the opponent has proposed at least some offers and if he is not dropping below the best opponent's offer.

The main acceptance criterion of the `LastOfferNegotiator` is that an offer is accepted if its utility is equal or above the agent's current aspiration level, or if it matches a proposal previously made by the agent itself. Throughout the negotiation, the agent keeps track of the maximum utility offered by the opponent. To prevent exploitation, the agent rejects any proposal that falls significantly below this maximum utility until the late phase of the negotiation.

In the end-game, the agent will accept an offer if it exceeds either the agent's absolute safety floor or a predefined utility margin above its reservation value. However, to prevent exploitation, two conditions have to be met: the opponent must have made a minimum amount of prior offers, and the current proposal must not represent a severe utility regression compared to the opponent's historical best offer.

2 Opponent Modelling

To predict human preference weights from a limited offer history, the `LastOfferNegotiator` utilizes an adapted version of the Conflict-Based Opponent Modeling (CBOM) proposed by Keskin et al. (2023). The assumption here is that the human negotiator will concede on its least preferred item first.

Unlike the original CBOM this version does not track issue-value orderings, focusing solely on deriving issue weights. The weights are initialized as the inverse of the agent weights and updated by counting the number of conflicts of the most recent opponent's offer against their entire offer history. The agent will then consider the issue ordering having the least conflicts and update its belief accordingly to refine its estimation of the opponent's weights.

3 LLM Integration & Text Responses

The LLM is decoupled from the decision-making process, thereby exerting no influence on acceptance decisions or offer generation. The mathematical decision engine and the CBOM make all strategic choices earlier. The LLM is solely responsible for generating textual responses that function as explanations and persuasions.

First messages, acceptance messages and termination messages are fixed templates. Only rejection messages are generated dynamically. When creating a rejection message, the first step is to select a negotiation argument based on the current state. The argument selection includes negotiation indicators like offer similarity, utility differences, concession, stagnation and remaining time. These indicators are then converted into argument categories or negotiation situations. These include the following: fairness, concessions dynamics, negotiation progress, similarity of offers and time driven considerations. A scoring heuristic is used to determine the most appropriate argument category. The negotiation indicators are used to assign scores to one or more argument categories, to select the best match. Each category contains multiple fixed template sentences, which are then randomly chosen. The selected argument is afterwards embedded in a structured prompt together with the generated offer and explicit style and rule instructions.

References

Keskin, M. O., Buzcu, B., & Aydoğan, R. (2023). Conflict-based negotiation strategy for human-agent negotiation. *Applied Intelligence*, *53*, 29741–29757. https://doi.org/10.1007/s10489-023-05001-9